

HC-01 — Climate-Informed Disease Outbreak Prediction

Organization	World Health Organization (WHO)
---------------------	---------------------------------

Problem Statement

Weather and climate conditions influence the incidence and geographical distribution of several diseases. WHO supports climate-informed disease early-warning approaches, but integrating environmental, epidemiological and population signals at local scale remains challenging.

Expected Solution

Develop a climate-health intelligence platform combining weather, satellite environmental observations, disease surveillance, population density, mobility, water availability and vector/environmental indicators. Generate localized outbreak probabilities weeks in advance and explain the environmental drivers.

Dataset / Reference

WorldPop
<https://www.worldpop.org/>

OpenDengue — A global dataset of publicly available dengue case count data
<https://www.nature.com/articles/s41597-024-03120-7>

HC-02 — Predicting Hospital Demand During Extreme Heat

Organization	NHS England
---------------------	-------------

Problem Statement

Extreme heat can increase demand for emergency and acute healthcare while simultaneously affecting infrastructure and staff availability. Health systems need to anticipate demand before hospitals become operationally overloaded.

Expected Solution

Develop a probabilistic forecasting system for emergency arrivals, heat-related admissions, ICU demand, beds, ambulances, staffing and critical medicines using weather, healthcare history, demographics, environmental conditions and hospital capacity. Identify facilities likely to exceed capacity.

Dataset / Reference

NHS England — A&E; Attendances and Emergency Admissions 2026-27

<https://www.england.nhs.uk/statistics/statistical-work-areas/ae-waiting-times-and-activity/ae-attendances-and-emergency-admissions-2026-27/>

HC-03 — Climate-Resilient Hospital Digital Twin

Organization	Mayo Clinic
---------------------	-------------

Problem Statement

Extreme weather can disrupt healthcare even when the hospital building remains intact. Flooding may affect access or electrical systems, heat can increase cooling requirements, and storms can disrupt utilities, staff availability and medical supply chains.

Expected Solution

Build a hospital digital twin that models weather, infrastructure, utilities, staff, supplies and patient flow. Simulate extreme-weather scenarios, identify critical failure points and recommend the optimal sequence of mitigation and resource-reallocation actions.

Dataset / Reference

PhysioNet — MIMIC-IV v3.1

<https://physionet.org/content/mimiciv/3.1/>

NHS England — Bed Availability and Occupancy

<https://www.england.nhs.uk/statistics/statistical-work-areas/bed-availability-and-occupancy/>

HC-04 — Hyperlocal Heat-Health Risk Prediction

Organization	Centers for Disease Control and Prevention (CDC)
---------------------	--

Problem Statement

Heat-health risk depends on more than temperature. Humidity, wet-bulb conditions, land cover, built environment, exposure duration and population vulnerability can substantially change risk. The challenge is to move from broad heat warnings toward dynamic, hyperlocal risk prediction.

Expected Solution

Develop a hyperlocal heat-health risk engine combining weather, humidity, satellite land-surface temperature, urban morphology, air quality and exposure patterns. Produce dynamic neighbourhood-level risk maps and identify locations where preventive interventions should be prioritized.

Dataset / Reference

Kaggle — Climate Change: Earth Surface Temperature Data

<https://www.kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data>

CDC/ATSDR — Heat & Health Index

<https://www.atsdr.cdc.gov/place-health/php/hhi/index.html>

HC-05 — Climate-Aware Emergency Response Network

Organization	International Federation of Red Cross and Red Crescent Societies (IFRC)
---------------------	---

Problem Statement

During floods, cyclones, heatwaves and other disasters, emergency systems must handle changing road accessibility, evolving hazards, increased demand and limited medical resources simultaneously.

Expected Solution

Develop an intelligent emergency-response system that continuously optimizes emergency vehicle positioning, evacuation routes, hospital selection and temporary medical resources using live hazard forecasts, road conditions, hospital capacity and emergency demand.

Dataset / Reference

Humanitarian Data Exchange (HDX)

<https://data.humdata.org/>

Humanitarian OpenStreetMap Team (HOT) on HDX

<https://data.humdata.org/organization/hot>